

1. Explain the nature of software and the fundamental principles of Software Engineering.
2. Explain the generic process model and different prescriptive process models.
3. Explain Agile Software Development and compare plan-driven development with agile development.
4. Explain Extreme Programming (XP) practices, testing in XP, and pair programming.
5. Explain Software Requirements Engineering and differentiate between user, system, functional, and non-functional requirements.
6. Explain Software Requirements Specification (SRS), its structure, and different ways of writing an SRS.
7. Explain requirements elicitation, analysis, validation, and requirements management.
8. Explain the software design process, design concepts, design models, and pattern-based software design.
9. Explain architectural design and component-level design with suitable examples.
10. Explain User Interface Design, UML diagrams, and different types of software testing.

MCQ

1. **Software Engineering is primarily concerned with:**
 - A) Hardware manufacturing
 - B) Systematic development and maintenance of software
 - C) Computer networking only
 - D) Database administration only
2. **Which of the following is a characteristic of software?**
 - A) It is manufactured like hardware
 - B) It does not wear out physically
 - C) It always fails due to physical damage
 - D) It is always inexpensive to develop
3. **Which of the following is a software engineering principle?**
 - A) Avoid documentation
 - B) Separation of concerns
 - C) Ignore requirements
 - D) Avoid testing
4. **Which process model follows a sequential development approach?**
 - A) Agile Model
 - B) Waterfall Model
 - C) Spiral Model
 - D) Concurrent Model
5. **RAD stands for:**
 - A) Rapid Application Development
 - B) Random Application Design
 - C) Requirement Analysis Development
 - D) Rapid Algorithm Design

6. **Which process model emphasizes the development of software in increments?**
 - A) Incremental Model
 - B) Waterfall Model
 - C) V-Model
 - D) Big Bang Model
7. **Agile development mainly emphasizes:**
 - A) Extensive documentation only
 - B) Customer collaboration and responding to change
 - C) Avoiding customer feedback
 - D) Fixed requirements throughout development
8. **Which of the following is an Agile methodology?**
 - A) Extreme Programming
 - B) Waterfall
 - C) Big Bang
 - D) V-Model
9. **In Extreme Programming, pair programming involves:**
 - A) Two teams working independently
 - B) Two programmers working together on the same task
 - C) Testing without coding
 - D) One programmer managing two projects
10. **JIRA is commonly used for:**
 - A) Image processing
 - B) Project and issue management
 - C) Operating system development
 - D) Hardware design
11. **A functional requirement specifies:**
 - A) What the system should do
 - B) The programming language only
 - C) The development cost only
 - D) The physical size of the computer
12. **Which of the following is a non-functional requirement?**
 - A) User login
 - B) Report generation
 - C) System performance
 - D) Order placement
13. **SRS stands for:**
 - A) Software Requirement Specification
 - B) System Resource Software
 - C) Software Resource System
 - D) System Requirement Structure
14. **Requirements validation is performed to ensure that requirements are:**
 - A) Expensive
 - B) Correct, complete, and consistent

- C) Difficult to understand
 - D) Written only by programmers
15. **Which activity involves collecting requirements from stakeholders?**
- A) Requirements elicitation
 - B) Coding
 - C) Deployment
 - D) Refactoring
16. **Which of the following is a software design concept?**
- A) Modularity
 - B) Compilation
 - C) Debugging only
 - D) Installation
17. **Architectural design primarily describes:**
- A) Detailed source code
 - B) Overall structure of a software system
 - C) User passwords
 - D) Test data only
18. **UML stands for:**
- A) Unified Modeling Language
 - B) Universal Machine Language
 - C) User Management Language
 - D) Unified Machine Logic
19. **Which UML diagram represents interactions between objects over time?**
- A) Class diagram
 - B) Sequence diagram
 - C) Use-case diagram
 - D) Deployment diagram
20. **Which testing technique examines individual software components?**
- A) System testing
 - B) Acceptance testing
 - C) Unit testing
 - D) Integration testing